REVISION WORKSHEET

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AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Practice Questions and Worked Answers

Grade 10 Advanced | Module 17 Lesson 2 | Diffraction

All 30 questions are followed by answers. Study the reasoning, formula, substitution and units. Equivalent scientifically correct wording is acceptable.

Useful equations and guidance: Single slit: $W \approx 2\lambda L/w$. Grating: $d \sin\theta = m\lambda$; $d = 1/(\text{line density})$. Use consistent units.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 50–57.

P01. Diffraction is most noticeable when an opening is:

A. Infinitely wide B. Much wider than every relevant scale
C. Comparable in size to the wavelength D. Made of a single specific colour

Answer

C. Comparable in size to the wavelength.

P02. For a single slit, the central bright maximum is generally:

A. A perfectly dark line B. Infinitely narrow
C. Always identical to a double-slit fringe D. Wider than neighbouring bright maxima

Answer

D. Wider than neighbouring bright maxima.

P03. Narrowing a single slit while keeping wavelength fixed makes the central maximum:

A. Wider B. Narrower
C. Disappear at all angles D. Have zero intensity necessarily

Answer

A. Wider.

P04. A diffraction grating contains:

A. Only one thick lens B. Many closely spaced parallel slits or grooves
C. Only one plane mirror D. No periodic structure

Answer

B. Many closely spaced parallel slits or grooves.

P05. A grating spectroscope separates light mainly by:

A. Changing every photon's frequency B. Absorbing only red light
C. Wavelength-dependent diffraction angles D. Making all wavelengths leave at one angle

Answer

C. Wavelength-dependent diffraction angles.

P06. A single slit has width 0.200 mm. With λ = 500 nm and screen distance 2 m, find the central bright-band width in millimetres. Use W = 2λL/w.

A. 10 mm B. 20 mm
C. 5 mm D. -10 mm

Answer

A. 10 mm. $W = 2(500 \times 10^{-9})(2)/(0.200 \times 10^{-3}) = 0.01$ m = 10 mm.

P07. A single-slit pattern has central width 20 mm at screen distance 2 m with λ = 500 nm. Find the slit width in millimetres.

A. 0.2 mm B. 0.1 mm
C. 0.05 mm D. -0.1 mm

Answer

B. 0.1 mm. $w = 2\lambda L/W = 2(500 \times 10^{-9})(2)/(0.02) = 0.0001$ m = 0.1 mm.

P08. A diffraction grating has spacing d = 2000 nm. For λ = 500 nm, calculate the angle of the first-order maximum.

A. 28.96 ° B. 7.239 °
C. 14.48 ° D. -14.48 °

Answer

C. 14.48 °. $d \sin\theta = m\lambda$; $\theta = \sin^{-1}(1 \times 500/2000) = 14.48°$.

P09. A grating has 300 lines per millimetre. Calculate its slit spacing in micrometres.

A. 6.667 µm B. 1.667 µm
C. -3.333 µm D. 3.333 µm

Answer

D. 3.333 µm. $d = 1/N = 1/300$ mm = 3.333 µm.

P10. A grating spacing is 2000 nm and the wavelength is 500 nm. Find the largest possible integer diffraction order.

A. 4 B. 8
C. 2 D. -4

Answer

A. 4. Because $\sin\theta \le 1$, $m \le d/\lambda = 2000/500 = 4$. Largest integer m = 4.

P11. Define diffraction.

Answer

The spreading of waves through openings or around edges, arising from superposition of wave contributions.

P12. What determines the single-slit central-maximum width in the small-angle approximation?

Answer

$W \approx 2\lambda L/w$, so it increases with wavelength and screen distance and decreases with slit width.

P13. Define diffraction order.

Answer

The integer m labelling a constructive grating maximum relative to the central order m = 0.

P14. Give an example of a reflection grating.

Answer

The closely spaced grooves of a DVD can act as a reflection grating and separate white light into colours.

P15. Why are higher grating orders limited?

Answer

The condition $\sin\theta = m\lambda/d$ cannot exceed 1 in magnitude, so only orders with $|m\lambda/d| \le 1$ are possible.

P16. A single slit has width 0.200 mm. With λ = 500 nm and screen distance 2.5 m, find the central bright-band width in millimetres. Use W = 2λL/w.

Answer

$W = 2(500 \times 10^{-9})(2.5)/(0.200 \times 10^{-3}) = 0.0125$ m = 12.5 mm.

P17. A single slit has width 0.200 mm. With λ = 500 nm and screen distance 3 m, find the central bright-band width in millimetres. Use W = 2λL/w.

Answer

$W = 2(500 \times 10^{-9})(3)/(0.200 \times 10^{-3}) = 0.015$ m = 15 mm.

P18. A single-slit pattern has central width 30 mm at screen distance 3 m with λ = 500 nm. Find the slit width in millimetres.

Answer

$w = 2\lambda L/W = 2(500 \times 10^{-9})(3)/(0.03) = 0.0001$ m = 0.1 mm.

P19. A single-slit pattern has central width 40 mm at screen distance 4 m with λ = 500 nm. Find the slit width in millimetres.

Answer

$w = 2\lambda L/W = 2(500 \times 10^{-9})(4)/(0.04) = 0.0001$ m = 0.1 mm.

P20. A diffraction grating has spacing d = 2000 nm. For λ = 550 nm, calculate the angle of the first-order maximum.

Answer

$d \sin\theta = m\lambda$; $\theta = \sin^{-1}(1 \times 550/2000) = 15.96°$.

P21. A diffraction grating has spacing d = 2000 nm. For λ = 600 nm, calculate the angle of the first-order maximum.

Answer

$d \sin\theta = m\lambda$; $\theta = \sin^{-1}(1 \times 600/2000) = 17.46°$.

P22. A grating has 350 lines per millimetre. Calculate its slit spacing in micrometres.

Answer

$d = 1/N = 1/350$ mm = 2.857 µm.

P23. A grating has 400 lines per millimetre. Calculate its slit spacing in micrometres.

Answer

$d = 1/N = 1/400$ mm = 2.5 µm.

P24. A grating spacing is 2000 nm and the wavelength is 550 nm. Find the largest possible integer diffraction order.

Answer

Because $\sin\theta \le 1$, $m \le d/\lambda = 2000/550 = 3.636$. Largest integer $m = 3$.

P25. A grating spacing is 2000 nm and the wavelength is 600 nm. Find the largest possible integer diffraction order.

Answer

Because $\sin\theta \le 1$, $m \le d/\lambda = 2000/600 = 3.333$. Largest integer $m = 3$.

P26. Compare single-slit diffraction of red and blue light with the same slit and screen distance.

Answer

Red light has the longer wavelength and therefore the wider central maximum because W is proportional to λ.

P27. If a single slit is widened to twice its initial width, predict the new central bright-band width.

Answer

The width halves under $W = 2\lambda L/w$, assuming the same wavelength and screen distance.

P28. For a grating, why are red maxima at larger first-order angles than blue maxima?

Answer

For fixed d and $m = 1$, $\sin\theta = \lambda/d$. A longer wavelength requires a larger angle within the allowed range.

P29. A student confuses grating line density with spacing. Explain the correct conversion.

Answer

Spacing is the reciprocal of line density. Convert the density to lines per metre before taking its reciprocal to obtain d in metres.

P30. Describe how a grating can help identify an unknown gas-discharge source.

Answer

Observe the discrete spectral lines and compare their wavelengths or angular positions with known reference spectra; each element has characteristic lines.